EE3621: Digital Signal Processing

Lecture 21: Quantization Effects in DSP Systems
(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Why Quantization Matters (ADC limitations, fixed-point vs floating-point).
- **05:00 – 12:00 (7 mins):** Uniform Quantization & Error Model (step size, representation levels, white noise assumption).
- **12:00 – 18:00 (6 mins):** SQNR Derivation ($6.02b + 1.76$ dB).
- **18:00 – 25:00 (7 mins):** Coefficient Quantization (FIR/IIR, sensitivity, direct vs cascade forms).
- **25:00 – 30:00 (5 mins):** Round-off Noise in Fixed-Point IIR (additive noise model, PSD, noise power).
- **30:00 – 35:00 (5 mins):** Limit Cycles & Overflow (granular, overflow, saturation vs two's complement).
- **35:00 – 40:00 (5 mins):** Checkpoints & Full Derivations.

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2 Why Quantization Matters

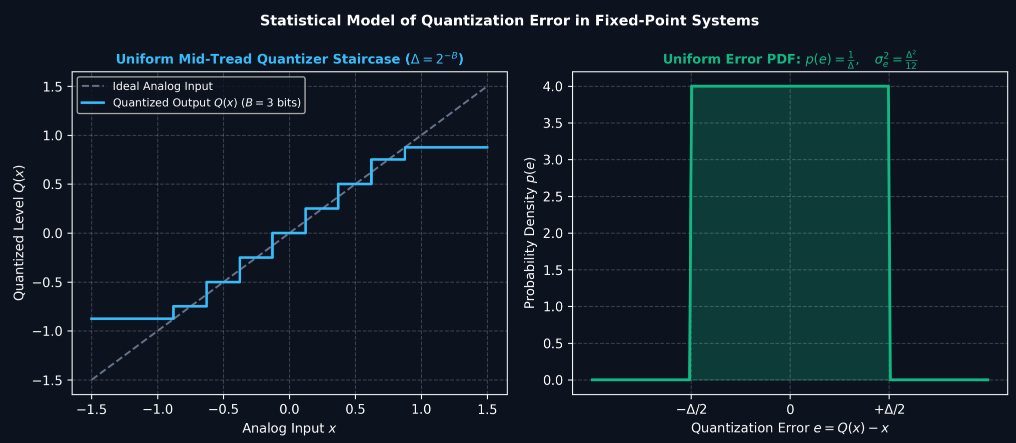

Figure 1: Uniform Mid-Tread Quantizer Characteristic and Uniform Error PDF Statistical Model ($\sigma_e^2 = \Delta^2/12$).

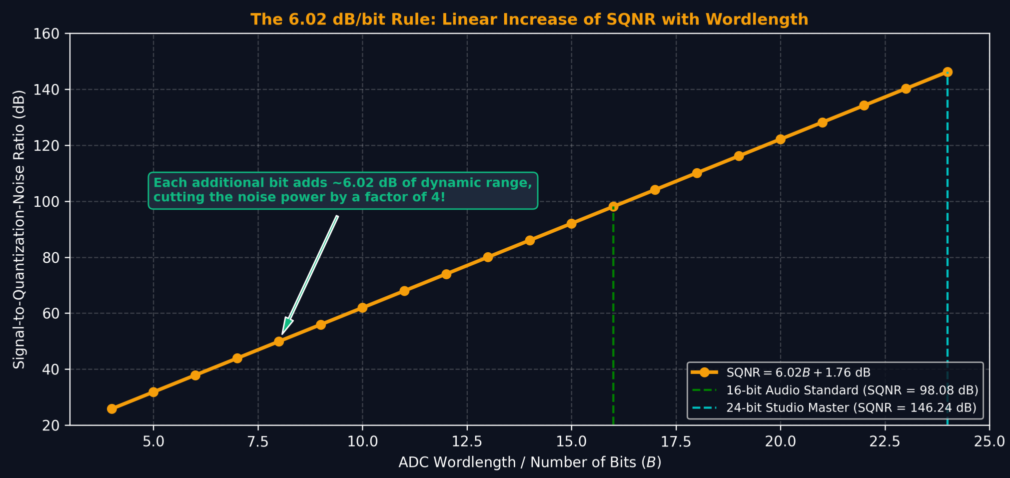

Figure 2: The 6.02 dB/bit Dynamic Range Rule: Linear SQNR Growth with ADC Wordlength.

In any practical Digital Signal Processing (DSP) system, we face finite word-length effects due to hardware limitations. An Analog-to-Digital Converter (ADC) maps continuous amplitudes into a finite set of digital levels, inherently discarding information. Fixed-point DSPs assign a strict number of bits to the integer and fractional parts. We discard bits after multiplication via truncation (chopping lower bits) or rounding (choosing the nearest representable value).

3 Uniform Quantization

Let the input signal range be from x_{min} to x_{max}. For a quantizer with b bits, the number of levels is $L = 2^b$. The step size Δ is:

$$\Delta = \frac{x_{max} - x_{min}}{2^b} \tag{1}$$

In the **Granular Noise Region**, the signal stays within bounds and error is constrained by Δ. In the **Overload Region**, the signal exceeds bounds and clipping causes severe distortion.

4 Quantization Error Model

The quantized signal $x_Q[n]$ is modeled as $x[n]$ plus additive noise $e[n]$:

$$x_Q[n] = x[n] + e[n] \quad \Rightarrow \quad e[n] = x_Q[n] - x[n] \tag{2}$$

Under the white noise model (Widrow assumption), $e[n]$ is uniformly distributed over $[-\Delta/2, \Delta/2]$. The variance is:

$$\sigma_e^2 = E[e^2] \tag{3}$$
$$= \int_{-\Delta/2}^{\Delta/2} e^2 \frac{1}{\Delta} de \tag{4}$$
$$= \frac{1}{\Delta}\left[\frac{e^3}{3}\right]_{-\Delta/2}^{\Delta/2} \tag{5}$$
$$= \frac{\Delta^2}{12} \tag{6}$$

5 Signal-to-Quantization-Noise Ratio (SQNR)

For a full-scale sinusoid $x[n] = A\cos(\omega n)$, peak-to-peak is $2A$. Thus, $\Delta = \frac{2A}{2^b}$.

$$P_x = \frac{A^2}{2} \tag{7}$$

$$P_e = \frac{\Delta^2}{12} = \frac{1}{12}\left(\frac{2A}{2^b}\right)^2 = \frac{A^2}{3 \cdot 2^{2b}} \tag{8}$$

$$\text{SQNR} = \frac{P_x}{P_e} = \frac{3}{2} \cdot 2^{2b} \tag{9}$$

$$\text{SQNR (dB)} = 10\log_{10}\left(\frac{3}{2} \cdot 2^{2b}\right) = 6.02b + 1.76 \tag{10}$$

Every additional bit increases SQNR by roughly 6 dB.

6 Coefficient Quantization in FIR/IIR

Filter coefficients h_k are quantized, perturbing poles and zeros:

$$\Delta H(e^{j\omega}) \approx \sum_k \frac{\partial H(e^{j\omega})}{\partial h_k} \Delta h_k \tag{11}$$

High-order direct form IIR filters are highly sensitive to coefficient quantization and can become unstable. Cascade forms (second-order sections) are much less sensitive since root dependency is decoupled.

7 Round-off Noise in Fixed-Point IIR

A b-bit times b-bit multiplication gives $2b$ bits, which is rounded. We model this as inserting white noise $e[n]$ after each multiplier. The output PSD is:

$$S_{ee}^{out}(e^{j\omega}) = \frac{\sigma_e^2}{2\pi} |H_{noise}(e^{j\omega})|^2 \tag{12}$$

Total output noise power is:

$$\sigma_{out}^2 = \sigma_e^2 \sum_{n=0}^{\infty} h_{noise}^2[n] \tag{13}$$

8 Limit Cycles

Fixed-point IIR filters can exhibit sustained oscillations for zero input.

- **Granular Limit Cycles:** Rounding/truncation inside the feedback loop.
- **Overflow Limit Cycles:** Occurs due to Two's Complement wraparound. Saturation arithmetic clips the value instead, heavily reducing this risk.

9 Floating-Point vs Fixed-Point

Fixed-point has constant absolute error. IEEE 754 Floating-point uses a mantissa and exponent ($x = M \cdot 2^E$), offering bounded relative error and a huge dynamic range. It is preferred when scaling is too complex or dynamic range fluctuates drastically.

10 Key Formulas Summary

Concept	Formula
Step Size	$\Delta = \frac{x_{max} - x_{min}}{2^b}$
Quantization Noise Variance	$\sigma_e^2 = \frac{\Delta^2}{12}$
SQNR for Full-Scale Sine	SQNR (dB) $= 6.02b + 1.76$
Output Noise Power	$\sigma_{out}^2 = \sigma_e^2 \sum_n h^2[n]$

Table 1: Key quantization formulas.

11 Checkpoint Questions

Q1: An 8-bit ADC has range ±5V. Calculate step size and quantization noise power.
A1: $\Delta = \frac{10}{256} \approx 0.039$ V. $\sigma_e^2 = \frac{(0.039)^2}{12} \approx 1.27 \times 10^{-4}$ W.

Q2: Derive the max amplitude A for a 12-bit ADC (range ±2.5V) to get SQNR = 70 dB.
A2: $\Delta = 5/4096 = 0.00122$ V. $P_e = \frac{\Delta^2}{12} = 1.24 \times 10^{-7}$ W. For 70 dB, $P_x/P_e = 10^7$. $P_x = 1.24$ W. $P_x = A^2/2 \Rightarrow A \approx 1.57$ V.

Q3: A first-order IIR $y[n] = x[n] + 0.8y[n-1]$ has multiplication noise $e[n]$. Total output noise power?
A3: $h_{noise}[n] = (0.8)^n u[n]$. $\sigma_{out}^2 = \sigma_e^2 \sum (0.8^{2n}) = \sigma_e^2 \sum (0.64)^n = \sigma_e^2 \frac{1}{1-0.64} \approx 2.78\sigma_e^2$.